Examining the Ability of Machine Learning Models to Predict SEP Events Utilizing Two Solar Cycles of Observations

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IAU General Assembly 2024

6-15 August 2024, Cape Town, South Africa

Introduction

Solar Energetic Particle Events (SEPs) can potentially harm modern civilization in several different ways:
- Degrade high frequency (HF) radio communications
- Incapacitate satellites
- Expose airline passengers to elevated radiation levels
- Endanger life in outer space

Many different kinds of models have been developed to predict SEP events:
- Physics-based
- Empirical
- Probabilistic
- Machine Learning-based

Leveraging big datasets, ML models are being trained to predict Solar Proton Events (SPEs) by considering diverse parameters, such as the magnetic field characteristics of solar Active Regions (ARs), the preceding soft X-ray and proton fluxes, and the statistics of solar radio bursts.

Previous Study / Research Goals

Our previous study [1] showed that:

- Using ML, Space-Weather MDI Active Region Patches (SMARP) data can correctly predict whether a solar flare will lead to a solar energetic particle event (SEP) 72% of the times
- The SMARP data set provides a leading time of 55.3 min for forecasting the SEP events
- One of the biggest limitations of Kasapis et al. 2022 was that only 65 SEP (positive) events were included in the dataset.

As in ML, the volume of data is one of the most important parameters for training a reliable prediction model. In this work we augment the existing training dataset and increase the number of positive instances (SEP events) by using the SHARP-SMARP dataset created by Kosovich et al. 2024 [2].

[1] Kasapis, S., et al. (2022). Interpretable machine learning to forecast SEP events for solar cycle 23. Space Weather, 20(2), p.e2021SW002842.
[2] Kosovich, P.A., et al. (2024). Time Series of Magnetic Field Parameters of Merged MDI and HMI Space-Weather Active Region Patches as Potential Tool for Solar Flare Forecasting. arXiv preprint arXiv:2401.05591.

SHARP-SMARP Dataset

Kosovich et al. (2024)

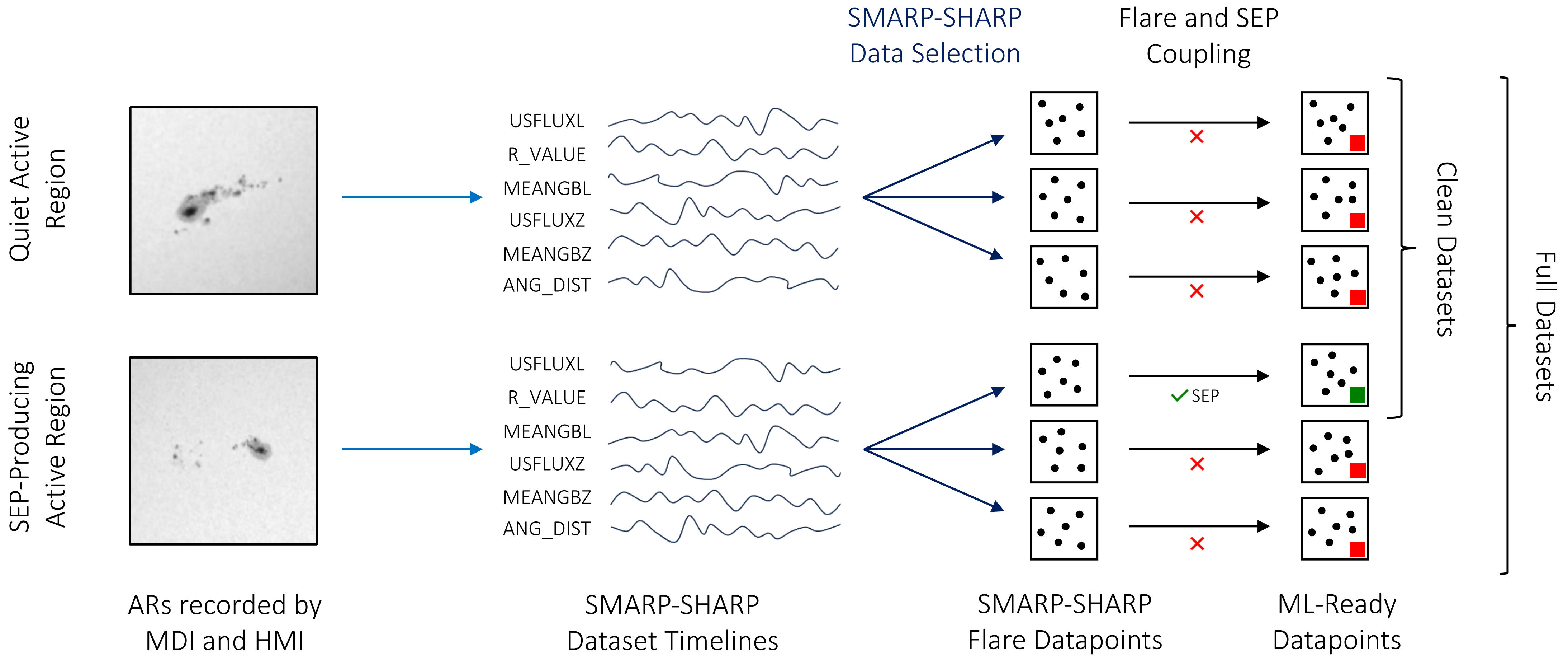
- Kosovich et al. (2024) combined SMARP (MDI) and SHARP (HMI) datasets into a single product for enhanced space weather forecasting.
- Filtering, rescaling, and merging of parameters, creating a uniform multivariate time series from April 4, 1996, to December 13, 2022.
- Identified time-lagged relationships between magnetic field properties of active regions and solar flare indices, suggesting predictive potential.
- The angular distance between an active region and Earth's magnetic foot point (assumed to be at W45°) is calculated using the SHARP-SMARP coordinates.

Keyword	Description
USFLUXL	**Total line-of-sight unsigned flux (Maxwells)**
R_VALUE	**Unsigned flux R near polarity inversion lines (Maxwells)**
MEANGBL	**Mean value of line-of-sight field gradient (Gauss/Mm)**
USFLUXZ	**Vertical component of the total unsigned flux (Maxwells)**
MEANGBZ	**Mean value of the vertical field gradient (Gauss/Mm)**
LAT_FWT	Stonyhurst latitude of flux-weighted center of active pixels (degrees)
CRLT_OBS	**Carrington latitude of the observer (degrees)**
LON_FWT	Stonyhurst longitude of flux-weighted center of active pixels (degrees)
CRLN_OBS	**Carrington longitude of the observer (degrees)**

[2] Kosovich, P.A., et al. (2024). Time Series of Magnetic Field Parameters of Merged MDI and HMI Space-Weather Active Region Patches as Potential Tool for Solar Flare Forecasting. arXiv preprint arXiv:2401.05591.

$$\text{dist} = \arccos\left(\sin(\theta_1)\sin(\theta_2) + \cos(\theta_1)\cos(\theta_2)\cos(\phi_1 - \phi_2)\right)$$

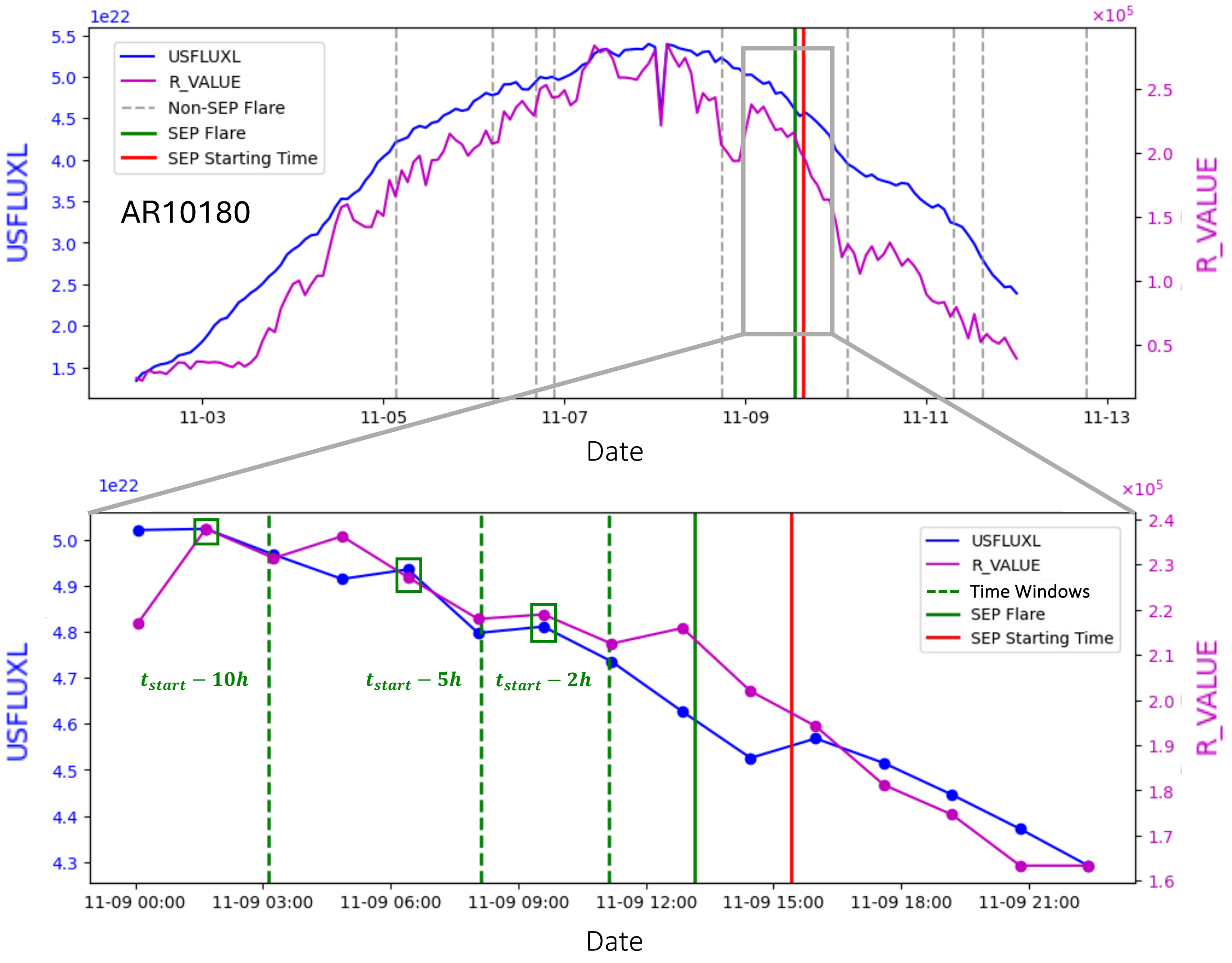
Dataset Creation



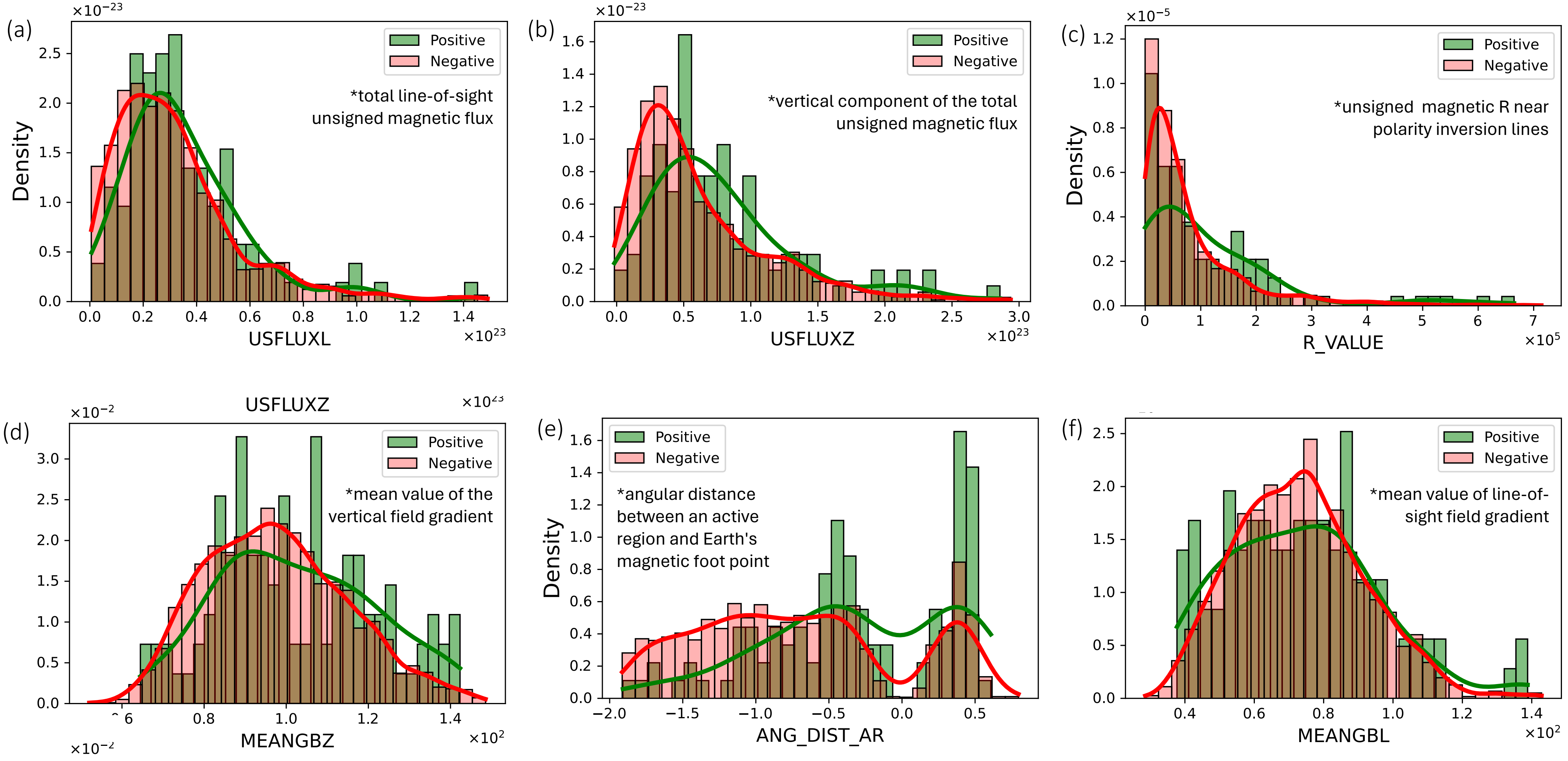
Data Selection

- The SMARP-SHARP dataset includes a continuous set of 21 keywords (only six of which are used here), representing homogeneous observations of active region patches from SoHO/MDI and SDO/HMI for the period between April 1996, and December 30, 2022

- Flare-based prediction: The final count for the flares that will be used for SEP prediction is **110 positive** (SEP productive) and **3246 negative** events (non-SEP productive).



Assessing the Keywords Predictive Abilities



Assessing the Keywords Predictive Abilities

Keyword	Regular		Clean		Difference
	Anderson	Smirnov	Anderson	Smirnov	Anderson / Smirnov
USFLUXL	2.3164	0.1309	6.8960	0.1813	+4.5796 / +0.0504
USFLUXZ	9.5532	0.2273	17.7214	0.2784	+8.1682 / +0.0511
R_VALUE	4.9641	0.1681	16.0888	0.2379	+11.1247 / +0.0698
MEANGBZ	3.8905	0.1407	9.4534	0.1885	+5.5629 / +0.0478
ANG_DIST	23.8579	0.3015	24.6131	0.3090	+0.7552 / +0.0075
MEANGBL	0.3643	0.0845	1.7362	0.1080	+1.3719 / +0.0235

- The Anderson-Darling statistic measures how well the data follow a particular distribution, with a focus on the tails.
- The Kolmogorov–Smirnov statistic quantifies the distance between the empirical distribution function of the sample and the cumulative distribution function of the reference distribution.

Dataset Variations

There are two types of settings tested in this work:

- **Regular / Clean:** By cleaning the dataset we omit the flares that did not produce an SEP from an SEP producing AR. This way the problem gets reduced from flare-centric to AR-centric, we predict whether an AR will produce an SEP rather than a flare.

- **Balanced / Imbalanced:** There are 3,356 negative samples in contrast to only 110 positive samples, an imbalance ratio of 1/30. Two types of training and validation schemes are used in our analysis: one where the positive and negative datasets are balanced and one where the on-to-thirty imbalance is retained.

	Clean	Regular
Balanced	Experimental	Semi Operational
Imbalance	Semi Operational	Operational

Results

We trained two types of ML models:

- **Support Vector Machines:** We train SVMs using four different filters, a linear, an RBF, a polynomial and a sigmoid function.

- **Regression Models:** We train two types of regression models, logistic and Ridge regression.

	SVM				Regression	
SMARP-SHARP Predictors	Linear	RBF	Polynomial	Sigmoid	Logistic	Ridge
ALL PREDICTORS	0.64 ± 0.10	0.62 ± 0.10	0.64 ± 0.10	0.62 ± 0.10	0.65 ± 0.09	0.65 ± 0.12
USFLUXZ, R_VALUE, ANG_DIST	0.66 ± 0.09	0.65 ± 0.10	0.66 ± 0.10	0.64 ± 0.10	0.65 ± 0.10	0.65 ± 0.09
USFLUXL, MEANGB, MEANGBZ	0.55 ± 0.10	0.55 ± 0.10	0.54 ± 0.10	0.53 ± 0.10	0.56 ± 0.10	0.56 ± 0.11
R_VALUE, ANG_DIST	0.67 ± 0.10	0.66 ± 0.10	0.67 ± 0.11	0.67 ± 0.09	0.66 ± 0.10	0.65 ± 0.10
USFLUXZ, ANG_DIST	0.66 ± 0.10	0.64 ± 0.11	0.64 ± 0.10	0.66 ± 0.10	0.66 ± 0.11	0.65 ± 0.10
USFLUXZ, R_VALUE	0.54 ± 0.11	0.58 ± 0.11	0.54 ± 0.08	0.58 ± 0.10	0.57 ± 0.09	0.55 ± 0.11

* All results presented are for the accuracy (ACC) metric.

Results

- We compare the SHARP-SMARP trained models results with those of Kasapis et al. (2022) and Lavasa et al. (2021).
- We use five different metrics to evaluate our models performance: the accuracy (ACC), the True Skill Score (TSS), the Heidke Skill Score (HSS), the False Alarm Rate (FAR) and the F1.

Setting	Training	Dataset	Model	ACC	TSS	HSS	FAR	F1
Operational	Imbalance	Regular	SVM Linear	0.56 ± 0.04	0.32 ± 0.10	0.05 ± 0.02	0.24 ± 0.10	0.71 ± 0.04
Kasapis et al. (2022)	Imbalance	Regular	SVM Poly	0.52 ± 0.05	0.01 ± 0.01	0.01 ± 0.01	0.98 ± 0.05	0.58 ± 0.06
Semi-Operational	Balanced	Regular	SVM Linear	0.66 ± 0.10	0.33 ± 0.19	0.33 ± 0.19	0.26 ± 0.14	0.63 ± 0.12
Kasapis et al. (2022)	Balanced	Regular	SVM RBF	0.65 ± 0.13	0.33 ± 0.28	0.31 ± 0.26	0.31 ± 0.10	0.75 ± 0.04
Semi-Operational	Imbalance	Clean	SVM Linear	0.67 ± 0.04	0.35 ± 0.13	0.08 ± 0.03	0.32 ± 0.12	0.79 ± 0.03
Experimental	Balanced	Clean	SVM Linear	0.69 ± 0.09	0.38 ± 0.19	0.38 ± 0.19	0.26 ± 0.14	0.67 ± 0.11
Experimental	Balanced	Clean	Logistic Reg	0.70 ± 0.09	0.39 ± 0.19	0.37 ± 0.19	0.30 ± 0.14	0.69 ± 0.10

- The Lavasa et al. (2021) "Flare-Noisy" Linear SVM Model with an Imbalanced data setting indicates generally better performance, as it exhibits better TSS (+0.14) and HSS (+0.36) scores, as it is trained on significantly more positive data.
- The current model demonstrates a significantly lower FAR than that reported by Lavasa et al. (2021), suggesting that it is less prone to incorrectly predicting SEP events.

Conclusions

- We utilize two Solar Cycles worth of data from the SOHO/MDI and the SDO/HMI instrument to train ML models in predicting SEP events.
- Statistical analysis shows that the vertical component of the total unsigned magnetic flux (USFLUXZ), the unsigned magnetic R near polarity inversion lines (R_VALUE) and the angular distance between an active region and Earth's magnetic foot point (ANG_DIST_AR) are the parameters related the most to the SEP prediction problem.
- When trained and tested in an operational setting, SVM and Regression models perform better in the presence of more data (two solar cycles instead of one).
- Although in an experimental setting the models ACC can reach 0.7, in an operational setting the accuracy decreases to 0.56, showing that the low dimensional SMARP-SHARP dataset can’t predict reliably an SEP.
- Future improvements in the problem of predicting SEP events using ML should be the use of high-dimensional data, deeper ML networks that can better capture the underlying dynamics of SEP production.

Thank you for your attention!

Kasapis et al. (2024)

Kasapis et al. (2022)

Kosovich et al. (2024)